

CSV202:SOURCE CODE MANAGEMENT LABORATORY

L:0 T:0 P:2 Credits:1

Course Outcomes: Through this course students should be able to

- CO1 :: Illustrate the installation and configuration of Git
- CO2 :: Identify the differences in various commits and working with Git branches
- CO3 :: Understand the use of Rollbacks in Git
- CO4 :: Analyze how to resolve Merge conflicts and identifying the usecase of rebase and revert
- CO5 :: Demonstrate the working of GitHub
- CO6 :: Understand the mechanism of cloning of GitHub repository

List of Practicals / Experiments:

Introducing Version Control System – Installing Git CLI on Windows, Mac & Linux

- Explain the Git Architecture & install Git on whatsoever OS participants are using. Ensure git commands are working & then configure Git.

Setting up a Git Repository & Initialization

- Explain the Repository Initialization & all the hidden artifacts involved in the process. Perform Reinitialization practical as well.

Working with Git – File States, Project Section, Workflow

- Perform the practical of tracking files through git, working with staging area, working with commit area. Different methods involved in working with commit & staging area. Explain commit id & tracking process of Git.

Working with Git History

- Work with git logs & its different variations to obtain different insights. Mention the importance of history in git.

Identifying Binary Files, Diffing Binary Files

- Explain Binary Files. Also perform the practical of getting the difference between various commits. Work with git diff command w.r.t various stages in git workflow.

Working with git branches

- Explain branches in git., explain all the ways using which different branches can be created. Explain the branch merging & its need as well.

Rollbacks in Git

- Explain the significance of HEAD, show the practicals for performing a rollback in timeline of git. Explain different types of resets in git.

Merge Conflicts Resolution in Git

- Explain Merge Conflicts, its source, & its solution practically with different strategies.

Rebasing & revert in Git

- Explain rebase, perform its practical, explain the difference between a merge & a rebase. Perform the practical for revert as well.

Introducing GitHub & Exploring GitHub

- Create a GitHub account & provide a tour of GitHub to the audience. Explain various elements of GitHub. Create a Repository on GitHub of public as well as private mode.

Cloning an existing Repository & remote setup

- Clone a GitHub Repository. Create a local Git repo & attach it to GitHub. Perform the complete remote setup.

Fork Existing Repository & open-source collaboration

- Perform the Repo Fork, explain the fork traversal, showcase star demo, issues demo to make the audience comfortable with the open-source work.

Text Books: 1. GITHUB FOR DUMMIES by PHIL HAACK SARAH GUTHALS, WILEY

References: 1. EFFECTIVE DEVOPS: BUILDING A CULTURE OF COLLABORATION, AFFINITY, AND TOOLING AT SCALE by JENNIFER DAVIS, O'REILLY